Analytics & Big Data

Session 2: Data preparation and exploration

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🎯 **Learning objectives**

- **Describe** variables, relationships between variables, and underlying structure in the data (e.g., clusters).
- **Prepare and explore** data sets using appropriate descriptive and visual techniques.
- **Explain** the role of exploratory data analysis in business decision-making.

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- Univariate EDA
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Foundations

Before building analytical models, analysts typically engage in **data preparation and exploratory data analysis (EDA)**.

These activities are **closely intertwined and often iterative**: we prepare and structure the data while simultaneously exploring it to understand its characteristics and potential issues.

Typical activities include:

- **Data preparation** (e.g., structuring, cleansing, integration)
- **Descriptive statistics** (e.g., mean, variance, frequencies)
- **Visualization techniques** (e.g., histograms, boxplots, scatterplots)

Goal: develop an initial understanding of the data, detect patterns or anomalies, and ensure that the data is suitable for further analysis or modeling.

Data preparation

Data types

Conducting EDA and representing the "real world" requires us to use appropriate data types. Typically, we distinguish four fundamental data types, known as measurement scales:

Qualitative description of objects

- Nominal (special case: binary)
- Ordinal

Quantitative description objects

- Interval
- Ratio

Qualitative: Nominal scale

- Values of a nominal (aka. categorical) attribute are symbols or names of things
- Each value represents some kind of category, code, or state.
- This scale assumes existence of a finite number of equivalency classes, where each class is named or labeled.
- The values do not have any meaningful order.

Examples:

- Hair color = {black, blond, brown, grey, red, white, …}
- Postal codes = {96047, 96048, …}

Special case of the nominal scale: Binary scale – also called Boolean

- Nominal attribute with only 2 states
- 0/FALSE typically means absence and 1/TRUE presence.

Examples:

- Smoking: 1 indicates patients in a trial that smoke, 0 otherwise.
- Purchase: 1 indicates that a person purchased a product, 0 otherwise.

Qualitative: Ordinal scale[1]

- A categorical attribute with values that have a meaningful order, but the magnitude between successive values is not known
- The ordinal scale is useful for registering subjective assessments and things that cannot be measured objectively (often used in surveys for ratings)

Values cannot be multiplied or added, even if the numbers belong to the same scale.

Relations between values:

- Transitivity: If A>B, B>C, then A>C,
- Symmetry: : If A>B, then B<A

Examples:

- School grades = {A < B < C < D < E < F} or {1 < 2 < 3 < 4 < 5 < 6}
- Places in a competition = {1st, 2nd, 3rd, 4th, …}

1. Also known as a “rank scale”.

Quantitative: Interval scale

- A numeric measureable attribute in the form of integer or real values. The distance between the numbers or units on this scale is equal over all levels of the scale. Values of the interval scale have no natural “zero” point.
- Invariant under a linear transformation $ax + b$, $a > 0$, $b \geq 0$
- Although the sum of two interval-scale measurements is not meaningful by itself, their average can be computed

Examples:

- Celsius scale of temperature (there is no natural zero, 0°C is arbitrarily defined), a Celsius value c can be linearly transformed to a Fahrenheit value f: $f = 9/5 * c + 32$
- Dates and time (e.g., conversion from Julian to Gregorian calendar is possible)

Quantitative: Ratio scale

- Numeric values with a meaningful zero point (e.g., “zero” means absence)
- All arithmetic rules and functions can be applied (addition, subtraction)

Examples:

- Money, person’s age, market share, quantities purchased, speed, …
- Kelvin (K) temperature (has a true zero-point (0 °K=− 273.15 °C): It is the point at which the particles that comprise matter have zero kinetic energy

Scales and mathematical operations

Due to the different mathematical properties, not all statistical measures can be computed on different measurement scales:

Scale	Mathematical operations possible	Mode	Frequency	Percentiles / Median	Mean / Variance / SD
Nominal	$=, \neq$	X	X		
Ordinal	$=, \neq, <, >$	X	X	X	
Interval	$=, \neq, <, >$, linear transformation	X	(X)	X	X
Ratio	$=, \neq, <, >, *, /, +, -$	X	(X)	X	X

(X) The frequency (relative / absolute) may be calculated for a numeric variable, but makes only sense when the number of possible values is low

Learning focus

No need to memorize the table.
Be able to **explain the scales and give examples of operations that are appropriate.**

Data quality and preparation

In business analytics, we rarely work with carefully curated datasets. Instead, data often comes from **multiple systems, processes, and teams**, where it is primarily created for **operational purposes rather than analysis**. Because these data sources are not always standardized or under the analyst's control, quality issues are common:

- **Missing values** (e.g., incomplete historical market data)
- **Outdated values** (e.g., customer address has changed)
- **Inconsistent formats or units** (e.g., dates stored as text, mixed currencies)
- **Duplicate records** (e.g., customers with multiple accounts)
- **Measurement or data entry errors** (e.g., human data entry, broken sensors)

Before conducting analysis, we must ensure that the **data is fit for use** (Strong et al., 1997).

> **Typical data preparation actions**
>
> 1. Data structuring
> 2. Data cleansing
> 3. Data integration
> 4. Data transformation

💡 In many real-world analytics projects, **data preparation and cleaning can take up to ~80% of the total effort.**

Data structuring: Example

There are many ways to structure the same dataset.

Option A (wide)

Region	Year	Sales_Q1	Sales_Q2	Sales_Q3	Sales_Q4
Europe	2023	120	140	160	150
Asia	2023	100	110	120	130

Option B (long)

Region	Year	Quarter	Sales
Europe	2023	Q1	120
Europe	2023	Q2	140
Europe	2023	Q3	160
Europe	2023	Q4	150
Asia	2023	Q1	100
Asia	2023	Q2	110
Asia	2023	Q3	120
Asia	2023	Q4	130

➡ Which would you select as an input for a data analytics tool?

→ Solution: **Option B**, because this is what **data analytics tools typically expect**.

Data structuring: Principles

Data structure *typically* expected by data analytics tools

> **Vocabulary**
>
> - A **dataset** is a collection of **values** (e.g., nominal, categorical, numeric).
> - Each value belongs to a **variable** and an **observation**
>
> **Expected data structure**
>
> - Each variable forms a column
> - Each observation forms a row
> - Each type of observational unit forms a table

💡 Data must be structured according to the expectations of the analytical tool being used. Reviewing the tool's documentation and example datasets can help clarify the required format.

Based on the Tidyverse and the work of Wickham (2014).

Data structuring: Example continued

Dataset `df` (wide)

Region	Year	Sales_Q1	Sales_Q2	Sales_Q3	Sales_Q4
Europe	2023	120	140	160	150
Asia	2023	100	110	120	130

Convert wide → long

```python
long_df = df.melt(
    id_vars=["Region", "Year"],
    var_name="Quarter",
    value_name="Sales"
)

long_df
```

Result:

Region	Year	Quarter	Sales
Europe	2023	Sales_Q1	120
Asia	2023	Sales_Q1	100
Europe	2023	Sales_Q2	140
Asia	2023	Sales_Q2	110
…	…	…	…

Data structuring: Example continued

As a last step, we need to clean the `Quarter` column by removing the `"Sales_"` prefix:

```
long_df["Quarter"] = long_df["Quarter"].str.replace("Sales_", "")
```

Final tidy dataset:

Region	Year	Quarter	Sales
Europe	2023	Q1	120
Europe	2023	Q2	140
Europe	2023	Q3	160
Europe	2023	Q4	150
Asia	2023	Q1	100
Asia	2023	Q2	110
Asia	2023	Q3	120
Asia	2023	Q4	130

Note: we will cover selected transformation operations like melt in the exercises.

Data cleansing

Data issue	Example	Data cleansing action
Missing values	Historical market data missing for some days	Impute values (e.g., interpolate prices), fill with mean/median, or mark as NA
Outdated values	Customer moved but old address still stored	Update records from CRM, external address verification service
Outliers / implausible values	Age = 250, salary = −5000	Detect and remove, cap values, or replace with plausible values
Inconsistent formats	Dates stored as `"03/01/23"` and `"2023-01-03"`	Standardize to a single format (e.g., ISO date `YYYY-MM-DD`)
Inconsistent units	Revenue in USD and EUR mixed in one column	Convert to common currency
Duplicate records	Same customer registered twice	Identify duplicates and merge records
Data entry errors	`"Munich"` vs `"Munch"`	Correct spelling using reference tables
Sensor errors	Temperature sensor reports `9999`	Remove invalid values or replace with interpolated measurements

Data integration

Organizations often store data in **different systems** (e.g., CRM, sales systems, web analytics).

Customer database (CRM)

Customer_ID	Name	Country	Email
101	Alice	Germany	alice@example.com
102	Bob	France	bob@example.com
103	Clara	Germany	clara@example.com

Sales transactions

Customer_ID	Order_ID	Revenue
101	5001	200
101	5002	150
103	5003	300

For analysis, we can **integrate these datasets** using the shared key `Customer_ID`.

```
integrated_data = customers.merge(sales, on="Customer_ID")
```

This creates a combined dataset that links customer information with sales.

Customer_ID	Name	Country	Email	Order_ID	Revenue
101	Alice	Germany	alice@example.com	5001	200
101	Alice	Germany	alice@example.com	5002	150
103	Clara	Germany	clara@example.com	5003	300

💡 For analytical purposes, **redundant values are often acceptable**. The goal is to organize data into a **tidy structure**, where **each row represents one observation** and all relevant variables are available for analysis.

Note: We will cover data integration in more detail in the next lecture.

Data transformation: Example

Often, the way data is **stored** determines which mathematical operations are possible.

Example: timestamps stored as strings

user_id	timestamp
101	“2025-03-12 14:23:05”
102	“2025-03-12 18:02:11”
103	“2025-03-13 09:15:42”

If treated as a **string**, we can only perform operations such as:

- equality checks ($=$ / $\neq$)
- simple filtering or sorting

This **severely limits analysis**.

Format conversions help transform raw data into representations suitable for EDA and analytical models:

Representation	Example	Possible analysis
Numeric timestamp	1741789385	time differences, time series
Hour of day	14	activity patterns during the day
Day of week	Wednesday	weekday vs. weekend behavior
Day of year	71	seasonal patterns

💡 Data preparation transforms raw data into **formats that allow meaningful mathematical operations and analysis**.

Univariate exploratory data analysis

Foundations

> People are not very good at looking at a column of numbers or a whole data table and then determining important characteristics of the data. EDA techniques have been devised as an aid in this situation.

The following forms of EDA are typically distinguished:

	Univariate	**Multivariate**
Non-graphical	mean, median frequencies	cross-tabs correlations
Graphical	histograms dot plots	scatter plots clustering

Descriptive statistics

How to describe data (not the data type)?

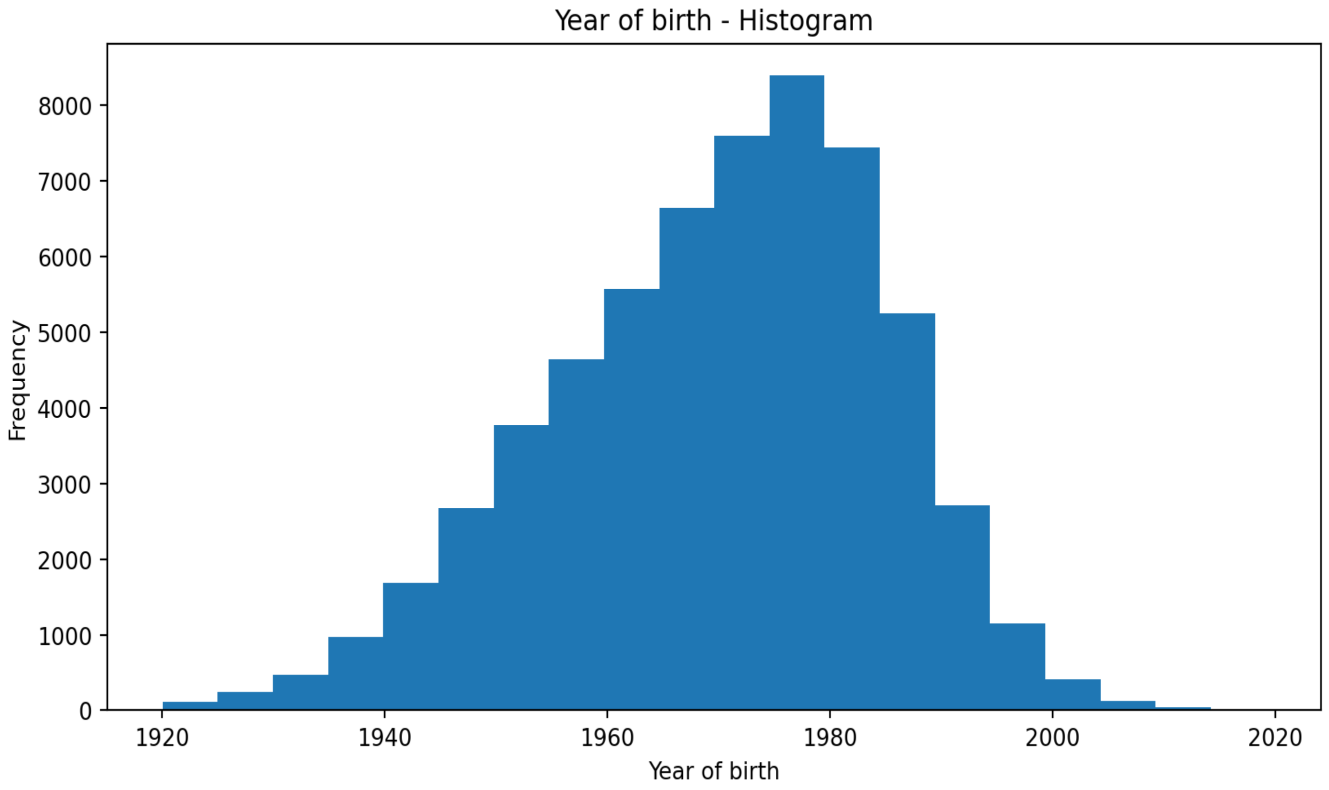
Customer ID	Name	Year of birth	Tariff
100216	Kevin Meyer	1983	A
271692	Lars Knopp	1963	B
892615	Anton Albert	1954	C
331625	Peter Pan	1988	D
...	...	...	...

What are the "key figures" of a distribution?

- Graphics work well for single variables, but for comparison of different variables and their distributions and later working with the data, numeric indicators are necessary
- Descriptive statistics provide us with numbers describing the characteristics of a distribution
- For qualitative / categorical data
  - **Mode**
  - **Relative frequency**
- For quantitative / numerical data
  - **Mean** (also known as **Expected Value**) – describing the position of the center
  - **Variance** (or **Standard Deviation**) – describing the spread
  - **Percentiles** (also known as quantiles / quintiles) – more detailed figures on the distribution

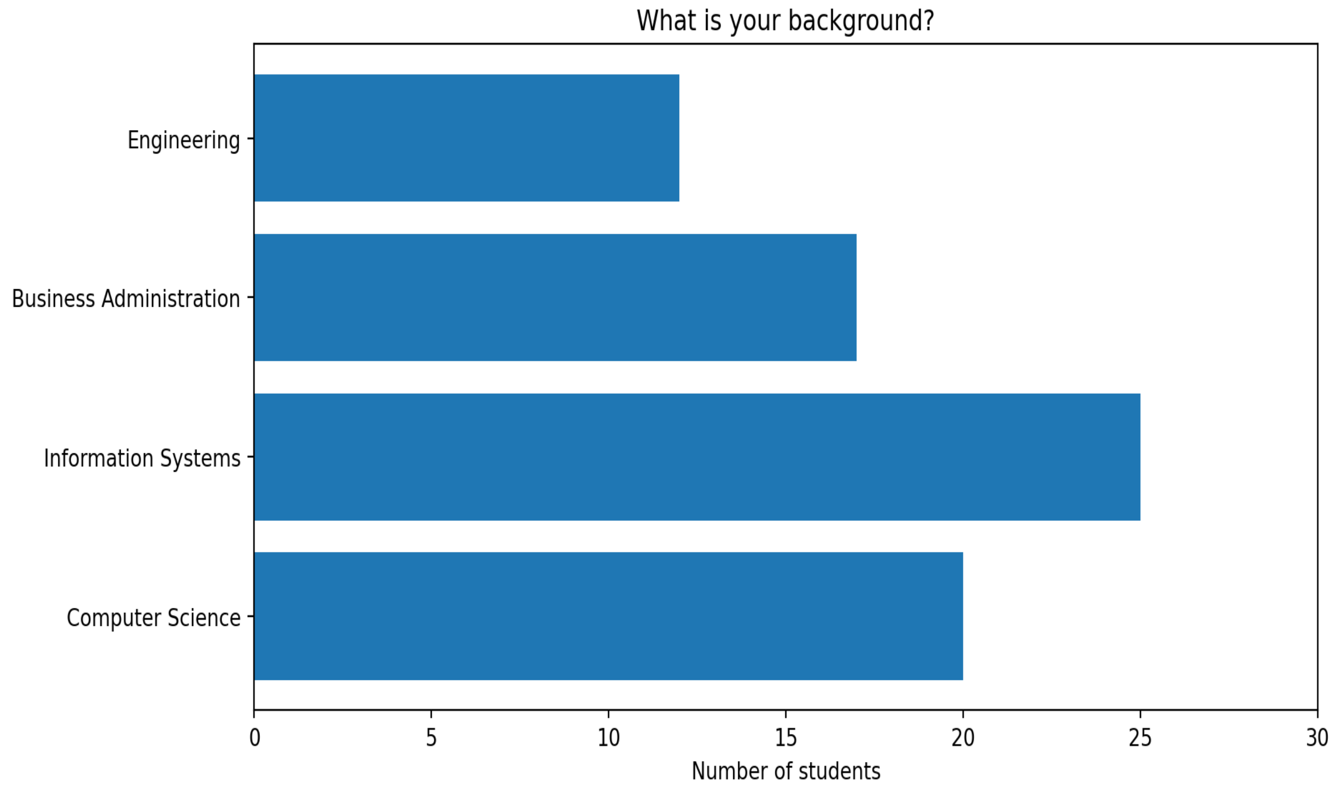
Describing data using descriptive statistics

- Basic statistical descriptions can be used to identify characteristics of the data and highlight which data values should be treated as noise or outliers.
- A good way to get an impression of continuous / numeric data is the **histogram**.
- The graph shows the range of observations on the horizontal axis, with a bar showing how many times each value occurred in the data set.



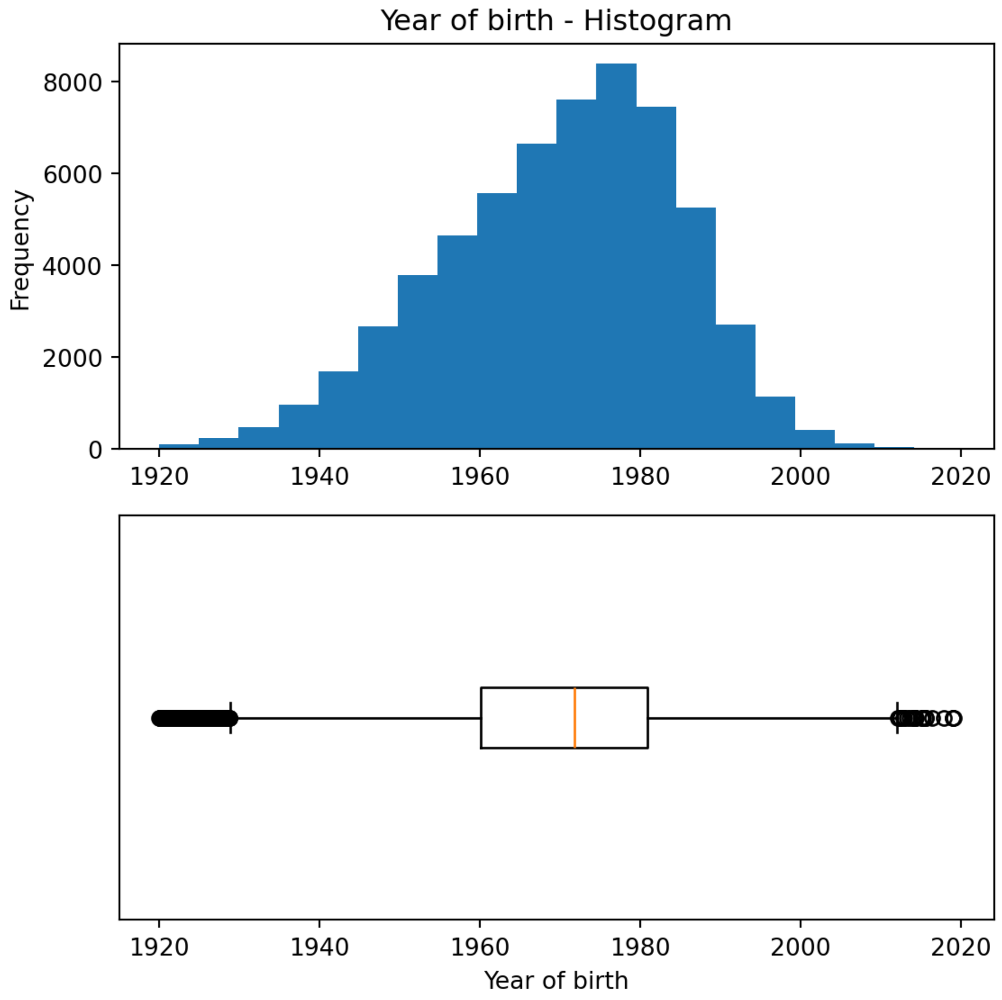
Mode

- The mode is the value that appears most in a set of data values
- The highest mode also has the highest relative frequency (see next slide)
- Example: On a party, you meet many other students, and you ask what they are studying.



Boxplot

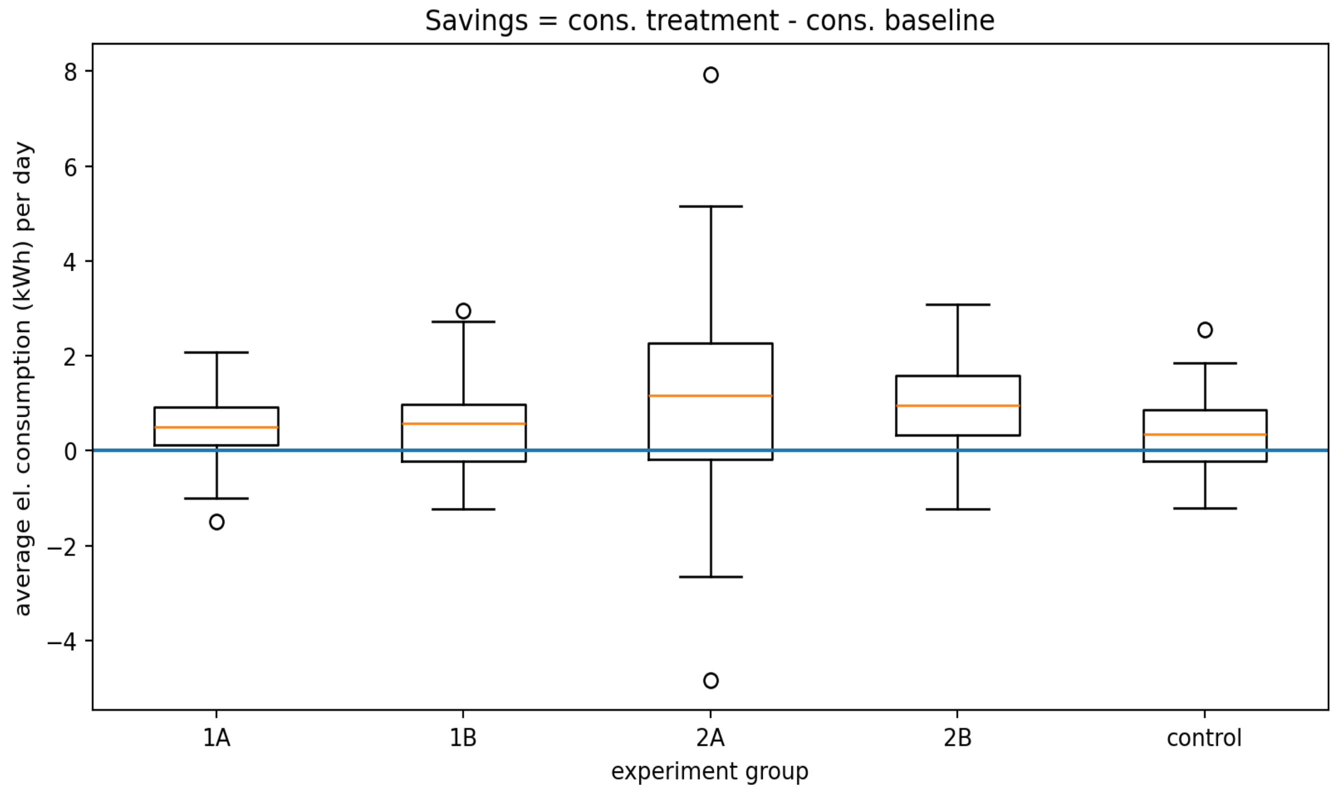
- The boxplot is a condensed illustration of a distribution
- It consists of
  - Median (thick line)
  - 25% / 75% percentiles
  - “Whiskers”: the quartiles $\pm\ 1.5 \times$ IQR
  - Outliers that are higher/lower than the whiskers



Boxplots can be used to compare different distributions

Example:

- In an experiment, households received different types of feedback on their electricity consumption
- The “control” group received no feedback
- The plot shows boxplots of the savings in electricity consumption after three weeks with feedback
- One can – for example – see that there is a higher spread in group 2A compared to the others



Multivariate exploratory data analysis

Multivariate non-graphical EDA

Multivariate non-graphical EDA techniques generally show the relationship between two or more variables in the form of either cross-tabulation for categorical variables or correlation statistics for numerical variables.

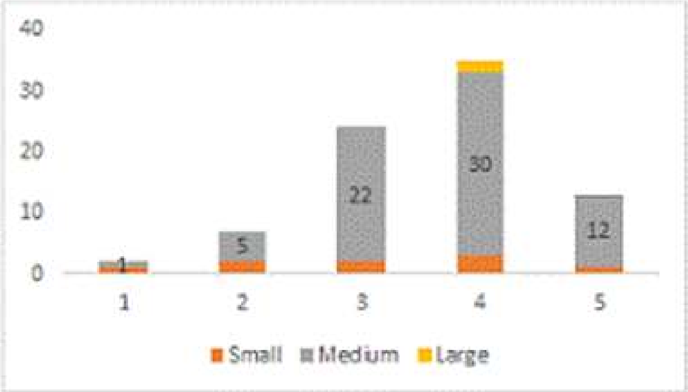
Frequency / Row Pct	Product Category 1	2	3	4	5	Total
Small	1 / 11.11	2 / 22.22	2 / 22.22	3 / 33.33	1 / 11.11	9
Medium	1 / 1.43	5 / 7.14	22 / 31.43	30 / 42.86	12 / 17.14	70
Large	0 / 0.00	0 / 0.00	0 / 0.00	2 / 100.00	0 / 0.00	2
Total	2	7	24	35	13	81

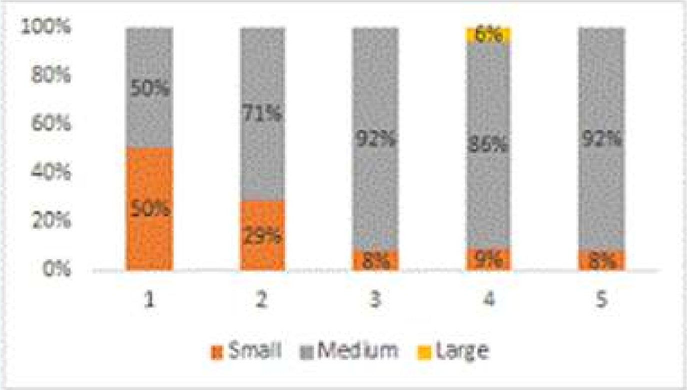
Frequency Missing = 77

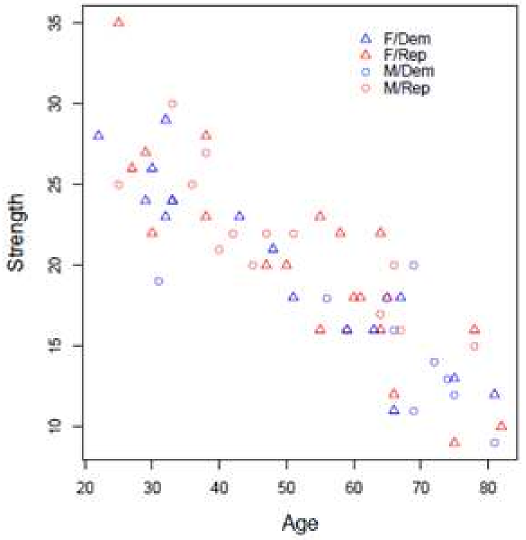
	V2	V3	V4	V5	V6
V2	-0.388				
V3	0.549	0.189			
V4	0.824	0.758	-0.619		
V5	0.802	0.833	-0.469	-0.946	
V6	-0.186	-0.659	0.349	0.555	0.649

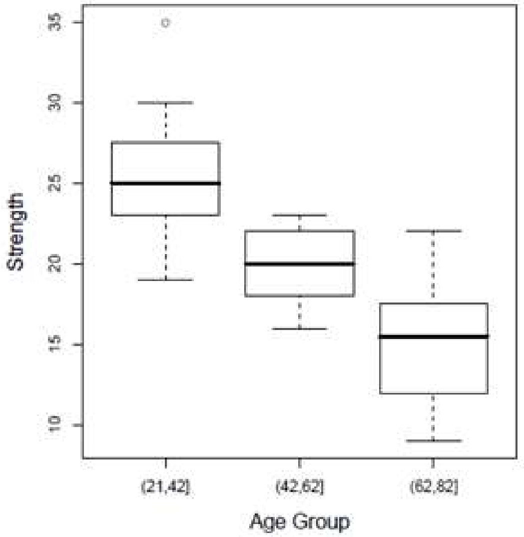
Multivariate graphical EDA

Multivariate graphical EDA techniques are scatterplots for numerical variables, Barcharts for categorical variables, or Boxplots for mixed types.









Bivariate statistics: Correlation

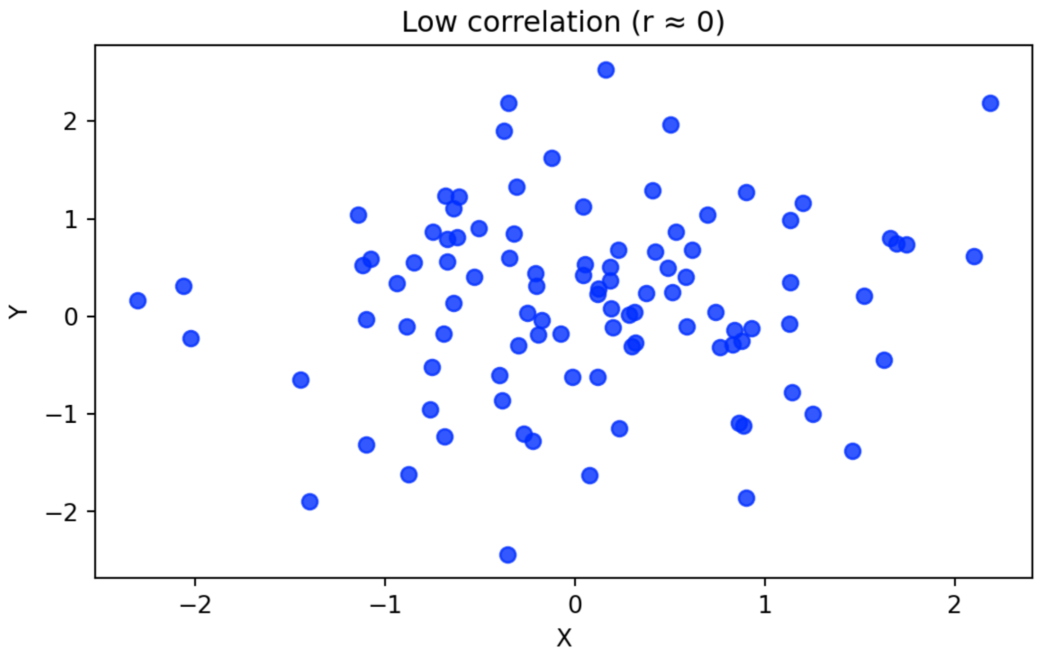
Correlation measures how strongly two variables move together. Suppose we observe two variables for the same observations:

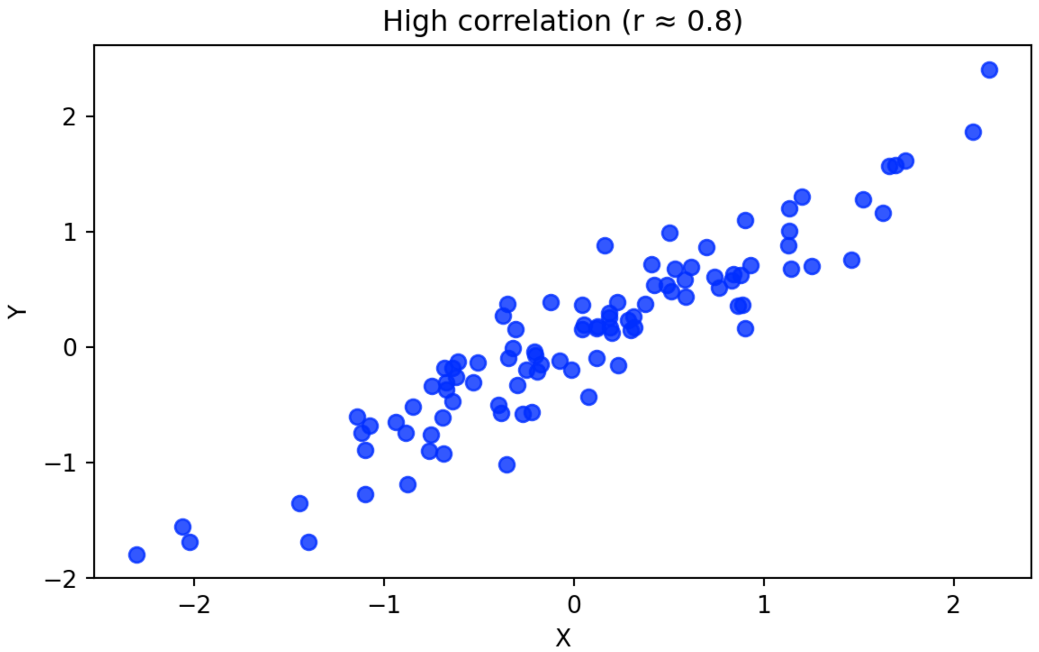
$$X = (x_1, x_2, \dots, x_n)$$

$$Y = (y_1, y_2, \dots, y_n)$$

The **Pearson correlation coefficient** is calculated as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

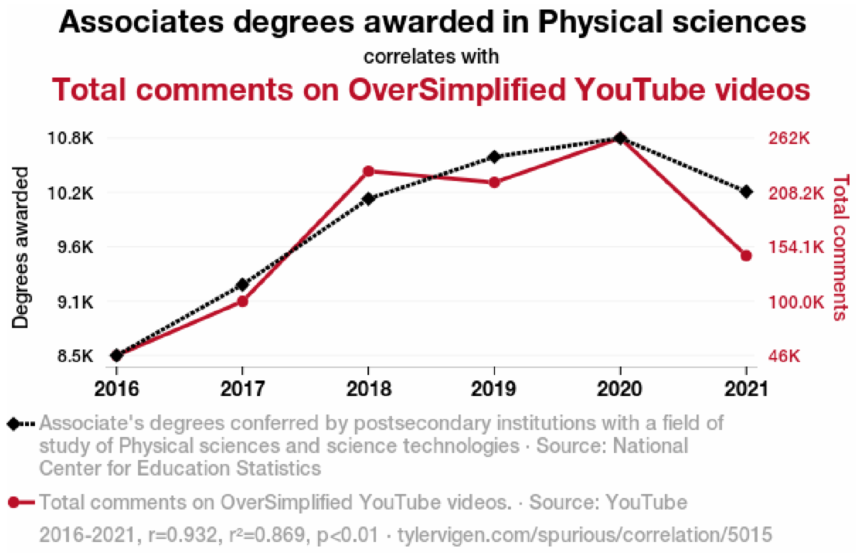




Correlation vs. causality

Correlation: Two data series behave "similar"

Causality: Principle of cause and effect



- Correlation does not imply causation.
- Bivariate correlations are often insufficient for prediction or explanation, requiring more powerful analytical models.
- If causal explanations are unavailable, decisions may need to rely on predictions.

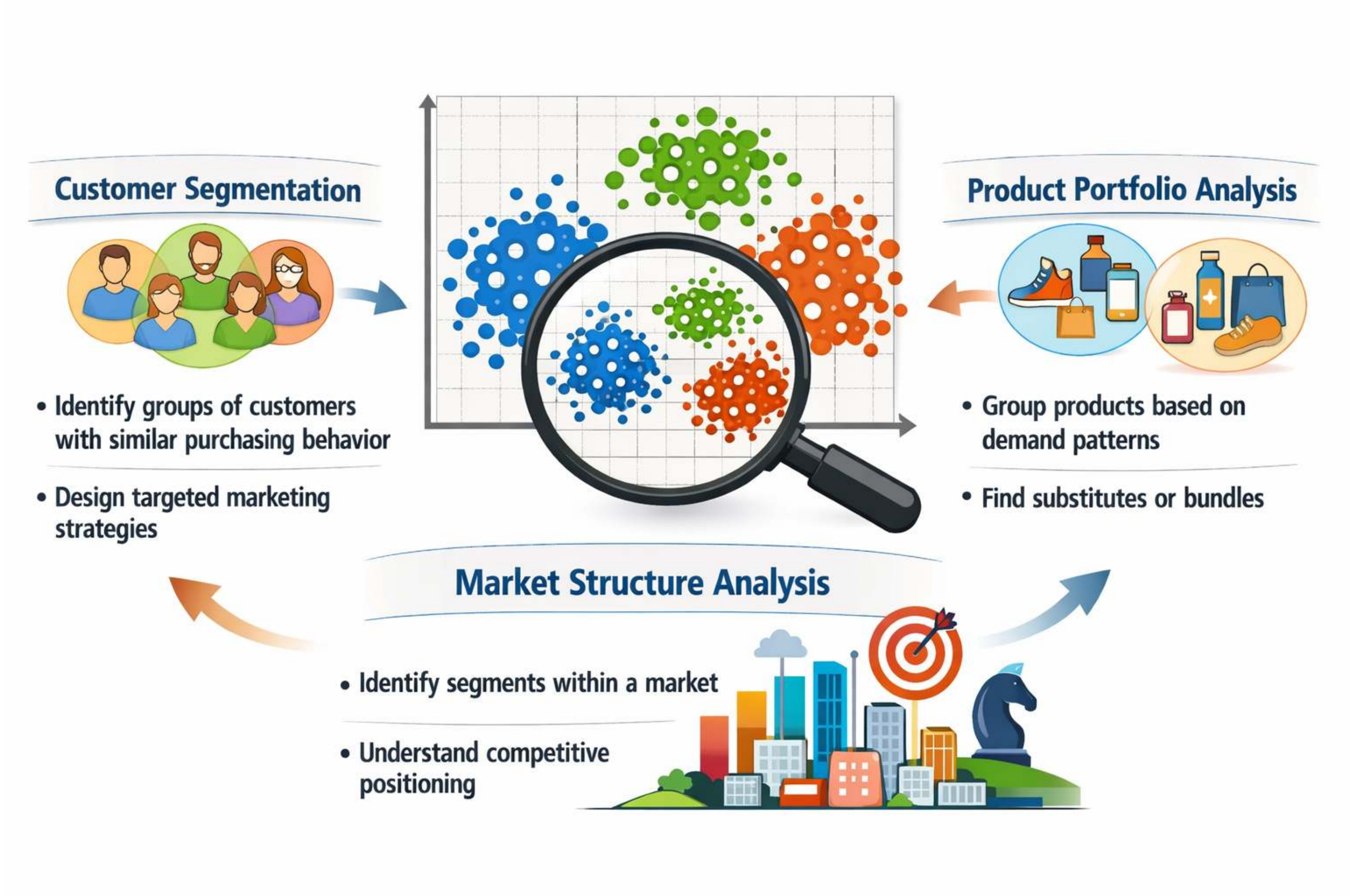
More examples are available at: https://www.tylervigen.com/spurious-correlations

Clustering in EDA

In **multivariate exploratory data analysis**, we often want to understand whether **observations form natural groups**.

Clustering is an EDA technique that groups **similar observations** based on their characteristics.

Common use cases include:



Families of clustering methods

Clustering algorithms can be grouped into several **families of approaches**.

Family	Examples	Key characteristics
Segmentation / Partitioning methods	**k-means**, k-medoids (PAM), bisecting k-means	divide observations into *k predefined clusters*; scalable for **large datasets**; widely used for **business/customer segmentation**
Hierarchical methods	**Agglomerative clustering**, divisive clustering	clusters formed through **successive merging or splitting**; produce **dendrograms (cluster trees)**; useful for **exploratory structure discovery**
Density-based methods	DBSCAN, HDBSCAN	clusters defined as **regions of high density**; identify **arbitrary shapes** and **outliers/noise**
Probabilistic / model-based methods	Gaussian mixture models	clusters modeled as **probability distributions**; allow **soft cluster membership** when groups overlap

In this lecture we focus on **k-means** (segmentation) and **agglomerative hierarchical clustering**.

Segmentation clustering

Segmentation methods divide observations into **k clusters**.

Goal:

- maximize similarity **within clusters**
- maximize differences **between clusters**

The most widely used algorithm:

k-means clustering

Key assumption:

clusters are roughly **spherical and similar in size**.

k-means algorithm

For a given **k**, the algorithm finds cluster centers that minimize **within-cluster sum of squares**.

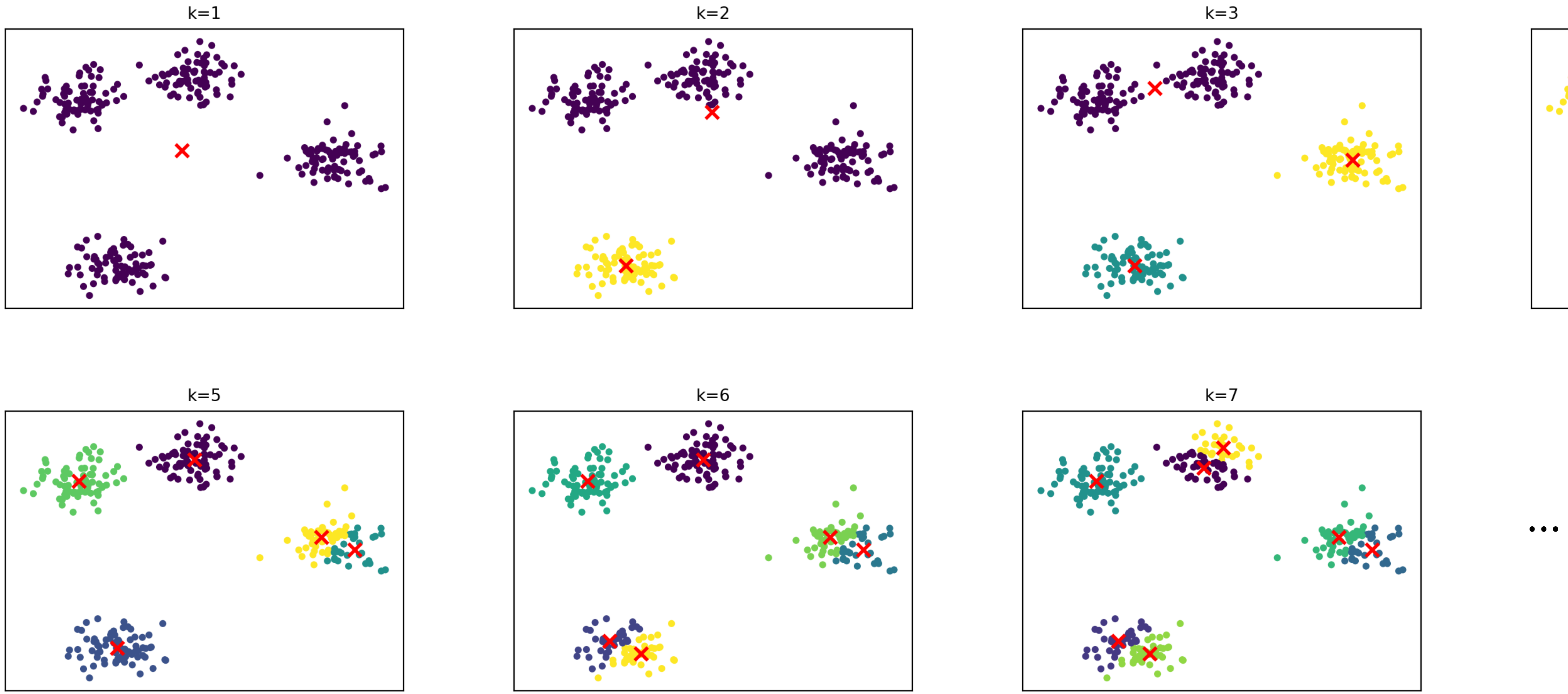
Steps:

```
Randomly assign data points to k clusters

Loop:

    calculate centroids for each cluster

    for each data point:
        compute distance to all centroids

        if x is not closest to its current centroid:
            assign x to the nearest centroid

    if no changes in centroids and cluster assignments:
        break
```

Strengths: fast and scalable, easy to interpret

Limitations: must choose **k**, sensitive to scaling and outliers.

How cluster solutions evolve



Note: The plots show the final result after the algorithm has converged, i.e., once observation assignments and cluster centroids no longer change.

k-means in Python

The following runs a k-means clustering analysis in Python:

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=k)
labels = kmeans.fit_predict(X)
```

💡 `KMeans(...)` + `fit_predict(X)` runs the iterative algorithm we saw earlier:

- initialize centroids, assign points to nearest centroid, update centroids (repeat until convergence)

It returns an array of cluster labels (corresponding to each observation). . . .

💡 Input data `X` must have a specific structure, as discussed in data preparation:

- rows = observations (data points)
- columns = features (dimensions)

Example:

```
X = [
    [1.2, 3.4],
    [1.0, 3.0],
    [5.5, 8.1],
]
```

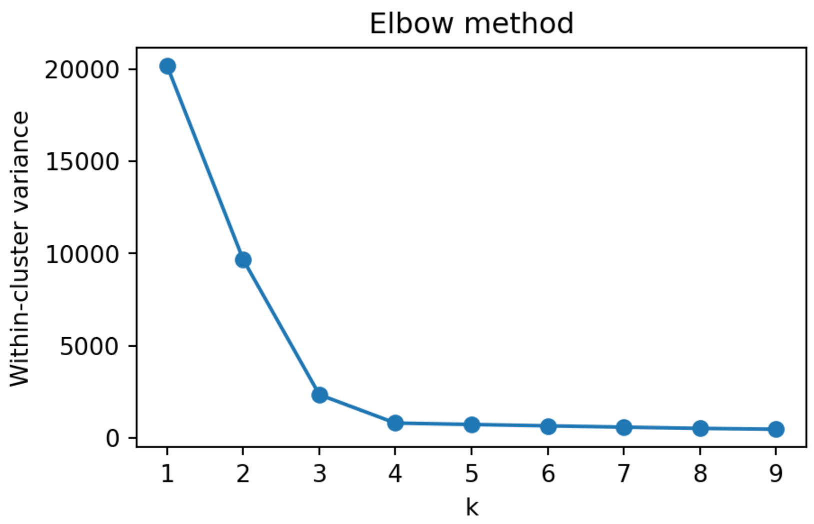
Choosing the number of clusters

A common heuristic is the **elbow method**.

Idea:

- increase k
- measure improvement in model fit
- look for point where improvement slows

Example metric: within-cluster variance (inertia)



Choose k at the **“elbow”** (diminishing returns)

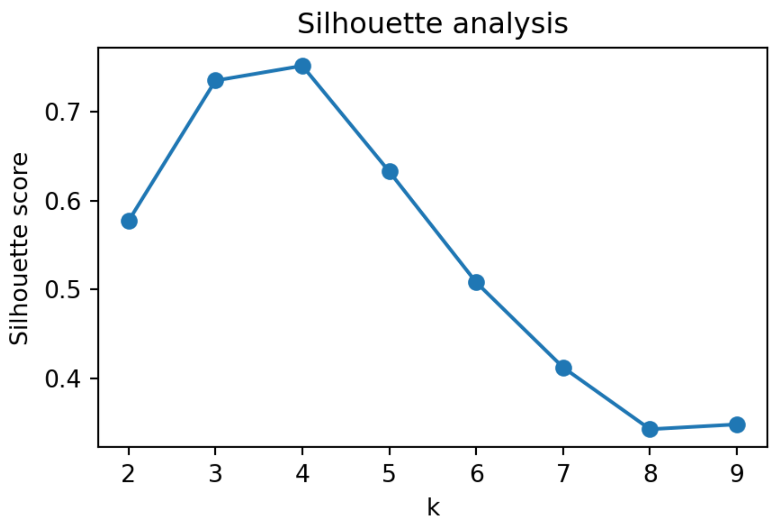
Choosing the number of clusters

An alternative is the **silhouette score**.

Idea:

- evaluate cohesion (within clusters)
- evaluate separation (between clusters)
- combine both into a single score

Choose k that **maximizes the score**



Higher = better separation and compactness

Hierarchical clustering

Hierarchical clustering builds a **tree of clusters**.

There are two types of hierarchical cluster methods:

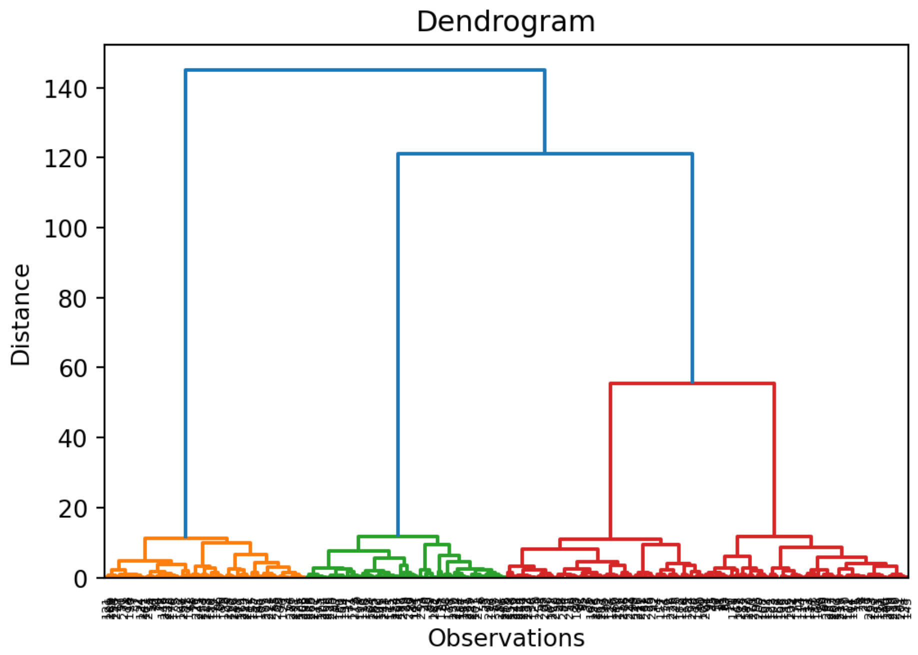
- **Agglomerative hierarchical clustering** is a bottom-up clustering method. It starts with every single data object in a single cluster. Then, in each iteration, it agglomerates (merges) the closest pair of clusters by satisfying some similarity criteria, until all of the data is in one cluster.
- **Divisive hierarchical clustering** is a top-down clustering method. It works in a similar way to agglomerative clustering but in the opposite direction. This method starts with a single cluster containing all data objects, and then successively splits resulting clusters until only clusters of individual data objects remain.

Dendrogram

A dendrogram is a tree diagram frequently used to illustrate the arrangement of the clusters produced by hierarchical clustering. The y-axis represents the value of this distance metric (e.g. euclidean distance) between the clusters.

In a dendrogram the widths of the horizontal lines give an impression about the dissimilarity of the merging object. Thus, a good cluster number might be at a point from where the width of the following horizontal lines is significantly smaller in length.

This allows us to explore clusters at **multiple levels of granularity**.



Measuring similarity between clusters

When merging clusters, we must define **distance between clusters**.

Common linkage choices:

- **single linkage**
  - nearest neighbor distance
- **complete linkage**
  - farthest neighbor distance
- **average linkage**
  - average pairwise distance
- **Ward linkage**
  - merges clusters minimizing variance increase.

Evaluation of clusters

⚠ Clustering algorithms will **always produce groups**, even when no meaningful structure exists.

Therefore clusters must be **evaluated**.

Common evaluation approaches include:

- **internal validation**
  - silhouette score
  - cluster stability
- **external validation**
  - compare with known labels or metadata
- **interpretability checks**
  - do cluster profiles make business sense?

Based on Balijepally et al. (2011).

Why clustering is useful in analytics workflows

Clustering often supports **later analytical steps**.

For example:

- **Customer segments in data warehouses**
  - segments stored as features for reporting or dashboards
- **Heterogeneity in regression models**
  - different segments may follow different behavioral patterns
  - subgroup analysis or separate models
- **Machine learning pipelines**
  - clustering can generate **features** or **latent groups**
- **Hypothesis generation and model development**
  - clusters can reveal **previously unnoticed patterns or structures**
  - these patterns may suggest **new hypotheses** about behavior or relationships in the data
  - clustering can therefore **inform model specification**, variable selection, or interaction terms

Summary

Exploratory Data Analysis (EDA) helps us understand data **before modeling**.

Key steps:

- **Prepare data**
  - ensure data quality (missing values, outliers, formats)
  - structure data appropriately (rows = observations, columns = features)
- **Describe variables**
  - univariate statistics (mean, variance, distributions)
  - visualization (histograms, boxplots)
- **Analyze relationships**
  - multivariate patterns (correlation, scatterplots)
  - ⚠ correlation ≠ causation
- **Identify structure**
  - segmentation (elbow method/silhouette score))
  - hierarchical clustering (dendogram)

💡 EDA is **not the final analysis**, but a **foundation** for model building, decision-making, and hypothesis generation.

Survey: Session 2

https://forms.gle/XfRmmVbf1yaBwrky9

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References

Balijepally, V., Mangalaraj, G., & Iyengar, K. (2011). Are we wielding this hammer correctly? A reflective review of the application of cluster analysis in information systems research. *Journal of the Association for Information Systems*, *12*(5), 1. https://doi.org/10.17705/1jais.00266

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